

THUMULA VAMSHIDHAR RAO

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SUMMARY

Python engineer who builds AI-assisted developer tooling along with the guardrails and analysis that make it trustworthy. Built a multi-agent HTTP service where a deterministic engine – not the model – decides every figure and citation, fenced by an outbound pipeline that detects identifier leaks by shape rather than by known value and treats planted instructions in untrusted text as findings for human review instead of commands; 397 tests, 95.38/96 on unseen graded data. Also built a rule-based classification and triage pipeline measured against a hand-labeled set with a confusion matrix and a written false-positive analysis, and root-caused a systematic defect across a 25,000-event stream by testing candidate invariants against reference-output mismatches – the same detect, classify, and prove loop that static analysis triage runs on. Day-to-day in Python, Linux, Git, Docker, and CI.

TECHNICAL SKILLS

Programming: Python (primary), JavaScript/TypeScript, SQL, Bash – comfortable picking up new languages fast

AI/LLM & Agents: Multi-agent systems (Agno, LangGraph), LLM API integration, typed tool/function calling, deterministic rule-based routing, RAG, prompt-injection handling, LLM evaluation harnesses

Security & Code Analysis: Guardrail & validation pipelines, PII/identifier masking, shape-based detection rules over source and free text, invariant-based defect detection, false-positive analysis, input validation & sanitization, JWT auth, rate limiting

Backend & Automation: REST API design (FastAPI, Flask, Express.js), ETL & automation pipelines, idempotency design, retry/backoff with budgets, contract validation (JSON Schema, Pydantic), CI/CD (GitHub Actions), pytest

Data & Infra: PostgreSQL, SQLite, SQLAlchemy, MySQL, Redis, Docker & Docker Compose, AWS (S3, EC2), Linux, Git

EXPERIENCE

AI/ML Intern – Ebani Tech Pvt. Ltd.

June 2024

Hyderabad (Onsite)

- Built Python automation and data-processing pipelines over CRM and user-interaction data, and exposed the derived outputs through backend APIs consumed by internal reporting tools – reducing a manual reporting step to a scheduled job.
- Integrated LLM capabilities into an internal workflow so non-technical teams could query performance metrics in natural language, owning the prompt design, API integration, and validation of model output before it reached users.
- Worked in a Linux environment with Git-based review, and wrote scripts to automate repetitive data validation and reporting steps for the team.

PROJECTS

Multi-Agent AI Service with Deterministic Guardrails | *Python, Agno, Docker, LLM APIs*

2026

- Built a seven-agent HTTP service in which a deterministic Python engine computes every figure and citation *before* any model call, so results stay correct and byte-for-byte reproducible even during a total LLM outage – proven by a suite of **397 tests** that runs with no network and no model.
- Designed a two-stage security barrier no code path can bypass: inbound scope binding plus identifier masking inside the data accessor, and an outbound pipeline (marker scrub → identifier scrub → citation-ownership check → advice screen → JSON-Schema contract validation) that matches sensitive values **by shape rather than by known value**, so it also catches identifiers the code has never seen and repairs a leak to its masked form instead of discarding the response.
- Handled adversarial instructions planted in untrusted record text by summarizing and flagging them for human review rather than executing them; replaced default SDK retries (7.2s on a single call) with budgeted backoff honoring **Retry-After**, holding **p95 latency at 6.98s** against a 20s target, and scored **95.38/96** on unseen evaluation data.

Consumer Research Pipeline – Automated Classification & Triage | *Python, SQLite* | (*GitHub*)

2026

- Built a config-driven ETL pipeline that crawls a public API with exponential backoff, caches every raw response to disk before parsing, and loads records into SQLite via **INSERT OR IGNORE** on a source-stable primary key – idempotent end to end and safe to re-run after a mid-stage crash (verified: 371 inserts, then 0 on a second full run).
- Classified every record with **auditable deterministic rules** (ordered keyword precedence) rather than an LLM, so any label is explainable from source, then measured the rules against a hand-labeled sample – accuracy, confusion matrix, and per-failure examples – and wrote up the two dominant failure modes (implicit sentiment with no trigger tokens, and taxonomy ambiguity between rule scope and human intent) as the basis for the next tuning pass.

- OpenEnv – Microservice Incident-Debugging Environment for AI Agents** | *Python, Docker* | (*GitHub*) 2026
- Built an HTTP environment server modeling real service dependencies (auth depends on database, payment on both) with three seeded failure modes including a latent cascading fault, plus a grading layer that scores an agent’s diagnosis and recovery – with strictly schema-constrained actions and observations so malformed agent output fails at the contract boundary instead of corrupting the run; finalist, Meta PyTorch OpenEnv Hackathon.
- Ledger Arena – Event-Sourced Engine with Invariant-Based Defect Detection** | *Python* 2026
- Root-caused an undisclosed systematic defect (a transaction reusing an already-settled `trade.id`) across a 25,000-event stream by intersecting reference-output mismatches with a battery of candidate invariants, and **discarded two of the strongest-looking invariants because they fired on valid input** – false-positive elimination before shipping a rule; structural idempotency (a seen-gate as the first statement of `apply()`, not per-handler bookkeeping) survived deliberate mid-run stream rewinds, backed by a 340-test suite.

EDUCATION

Vellore Institute of Technology	Vellore, Tamil Nadu
<i>Bachelor of Technology — GPA: 8.73 / 10.00</i>	<i>Sep 2022 – May 2026</i>

ACHIEVEMENTS & EXTRACURRICULAR

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- Finalist – Meta PyTorch OpenEnv Hackathon × Scaler School of Technology** 2026
- Selected as a finalist for the autonomous-SRE evaluation environment above. Also a finalist at the **National Entrepreneurship Challenge, IIT Bombay** (Feb 2024), presenting a technical product to an industry panel – experience explaining engineering work clearly to non-specialist audiences.